

# A Level Mechanics

Notes · Edexcel A Level Mathematics — Mechanics · Chapter 1: Modelling in Mechanics

## Chapter 1: Modelling in Mechanics

### Why this chapter matters

Mechanics applies mathematics to real physical situations, but real situations are messy: objects have size and shape, motion is affected by air resistance and friction, and countless other factors could in principle be taken into account. Before any calculation can begin, a real scenario has to be simplified into a mathematical model that keeps the physics that matters and strips out the physics that doesn't.

This chapter sets out the standard modelling assumptions used throughout the rest of the course, together with the vocabulary for describing quantities as scalars or vectors. Both ideas are used — and examined — in every chapter that follows, so it is worth getting comfortable with them now, before moving on to kinematics in Chapter 2.

### 1.1 Mathematical models and modelling assumptions

#### Concept

A **mathematical model** is a simplified representation of a real situation, built so that the mathematics of the course (Newton's laws, the kinematics equations, and so on) can actually be applied to it. Every model relies on **modelling assumptions** — stated simplifications that ignore effects considered too small to matter for the problem at hand.

Modelling assumptions are not “cheating”: they are a deliberate trade-off between realism and mathematical tractability. Without them, almost every real situation would be far too complicated to solve with the tools available at this level. For this reason, questions frequently ask you to *state* an assumption being used, or to *comment* on how realistic a given model is.

#### Key vocabulary

| Term                | Meaning                                                                                                                |
|---------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Particle</b>     | An object modelled as a single point with no size or shape, so rotation and air resistance due to shape can be ignored |
| <b>Rod</b>          | A rigid, straight object that does not bend or stretch                                                                 |
| <b>Light</b>        | Having no mass (used for strings, rods, springs and pulleys, so their own weight can be ignored)                       |
| <b>Inextensible</b> | Does not stretch (used for strings, so its length stays constant)                                                      |
| <b>Smooth</b>       | Frictionless (a smooth surface or pulley exerts no resistive frictional force)                                         |

|                |                                                                                                                   |
|----------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Uniform</b> | Having its mass evenly distributed, so its weight can be treated as acting at a single point (its centre of mass) |
|----------------|-------------------------------------------------------------------------------------------------------------------|

### Procedure

1. Identify which real-world complications are present in the scenario — for example size and shape, air resistance, friction, the mass of a string or pulley, or stretching.
2. Decide which of these can reasonably be ignored for the situation described, and state the corresponding modelling assumption using the standard vocabulary above (for example, “modelled as a particle” or “the string is light and inextensible”).
3. When asked to comment on a model, explain specifically what the assumption ignores, and whether that omission is likely to matter in this particular context. For example, air resistance is often reasonable to ignore for a dense, compact object moving slowly, but not for a light, wide object moving quickly.

### Worked example

#### EXAMPLE 1.1

**Question.** A skydiver falls from an aeroplane. Initially, the skydiver is modelled as a particle. (a) State one modelling assumption implied by treating the skydiver as a particle, and what it ignores. (b) After the parachute opens, explain why the particle model is no longer appropriate.

**Step 1 (part a) — state what “particle” ignores.** Modelling the skydiver as a particle means their size and shape are ignored. In particular, this means that air resistance caused by the shape and orientation of their body is not taken into account. **B1**

**Step 2 (part b) — check whether the assumption still holds.** Once the parachute opens, air resistance becomes a major force acting on the system, and the large surface area and shape of the parachute become essential to the motion. These effects can no longer be reasonably ignored, so the particle model is no longer a good representation of the situation. **B1**

**Answer:** (a) The particle model ignores the skydiver's size and shape, and therefore air resistance due to shape/orientation. (b) After the parachute opens, air resistance and the parachute's shape become significant to the motion, so modelling the system as a particle is no longer appropriate.

### Exam tip

#### EXAM TIP

When a question asks you to “criticise” or “comment on” a model, a strong answer always names the specific assumption and explains its effect on the particular scenario given, rather than making a generic statement such as “the model isn't realistic.” A useful habit is to finish every such sentence with “because...”, tying the assumption directly back to the context given in the question — this single habit is worth more exam marks than any amount of extra detail.

## 1.2 Quantities, units, and scalars vs vectors

### Concept

Every quantity in mechanics is either a **scalar** (fully described by its magnitude alone — for example distance, speed, time, mass) or a **vector** (needing both magnitude and direction — for example displacement, velocity, acceleration, force).

Confusing the two is one of the most common sources of error in this subject. Speed and velocity, or distance and displacement, are *not* interchangeable: only the vector version carries direction information, and losing that direction can turn an otherwise correct method into a wrong final answer.

All mechanics calculations use SI units: metres (m) for length, kilograms (kg) for mass, seconds (s) for time, and Newtons (N) for force. Any quantities given in other units must be converted before you start a calculation.

### Key vocabulary

| Term                | Meaning                                                                                         |
|---------------------|-------------------------------------------------------------------------------------------------|
| <b>Scalar</b>       | A quantity with magnitude only (e.g. distance, speed, mass, time)                               |
| <b>Vector</b>       | A quantity with both magnitude and direction (e.g. displacement, velocity, acceleration, force) |
| <b>Displacement</b> | A vector: the straight-line distance and direction from a starting point                        |
| <b>Distance</b>     | A scalar: the total length of path travelled, regardless of direction                           |

### Key formula

#### KEY FORMULA

$$W = mg$$

where  $W$  is weight (in Newtons, N),  $m$  is mass (in kilograms, kg), and  $g$  is the acceleration due to gravity (usually taken as  $9.8 \text{ m s}^{-2}$  unless the question states otherwise). Because weight is a force, it is a vector — for an object resting on or falling towards the Earth, it acts vertically downwards.

### Procedure

1. Identify whether each given or required quantity is a scalar or a vector, by checking whether direction is relevant to the problem.
2. Convert any quantities that are not already in SI base units (for example km/h to m/s, or grams to kilograms) before starting a calculation.
3. When working with a vector quantity, establish a clear positive direction first — for example, “taking upwards as positive” — and state it explicitly. This sign convention must then be used consistently throughout the solution.
4. Use  $W = mg$  to convert between mass (a scalar) and weight (a vector force). Remember that mass and weight are not the same quantity, even though they are often used interchangeably in everyday language.

## Worked example

### EXAMPLE 1.2

**Question.** (a) State whether each of the following is a scalar or a vector: (i) the speed of a car, (ii) the acceleration of a falling ball, (iii) the mass of a box. (b) A car travels at a constant speed of 54 km/h. Find its speed in m/s. (c) A particle has mass 2 kg. Find its weight, taking  $g = 9.8 \text{ m s}^{-2}$ .

**Step 1 (part a) — classify each quantity.** (i) Speed is a scalar, because no direction is specified. (ii) Acceleration is a vector, because it has a direction (for example, downwards for a falling ball). (iii) Mass is a scalar. **B1 B1 B1**

**Step 2 (part b) — convert km/h to m/s.** To convert kilometres per hour to metres per second, multiply by 1000 (the number of metres in a kilometre) and then divide by 3600 (the number of seconds in an hour). **M1**

$$54 \text{ km/h} = 54 \times 1000 \text{ m/h} = 54\,000 \text{ m/h}$$

$$54\,000 \text{ m/h} \div 3600 = 15 \text{ m/s} \quad \mathbf{A1}$$

**Step 3 (part c) — substitute into  $W = mg$ .** Substitute  $m = 2$  and  $g = 9.8$  into the formula. **M1**

$$W = mg = 2 \times 9.8$$

$$W = 19.6 \text{ N} \quad \mathbf{A1}$$

**Answer:** (a) (i) scalar, (ii) vector, (iii) scalar. (b) 15 m/s. (c) 19.6 N.

## Exam tip

### EXAM TIP

Mass and weight are often used interchangeably in everyday speech, but in mechanics they are distinct quantities. Mass is a scalar, measured in kilograms, and does not change with location. Weight is a vector force, measured in Newtons, and depends on the local gravitational field strength — this is why an object's weight would be different on the Moon even though its mass stays exactly the same. It is worth mastering the formula  $W = mg$  early, since it underpins force diagrams in every later chapter.

## Chapter summary

| Idea                         | Key point                                                                                                            |
|------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Modelling assumptions</b> | Simplifications (particle, light, inextensible, smooth, uniform) that make a real situation mathematically tractable |
| <b>Particle</b>              | No size or shape — ignores rotation and shape-dependent air resistance                                               |
| <b>Light / inextensible</b>  | No mass / does not stretch (used for strings, rods, pulleys)                                                         |
| <b>Smooth</b>                | Frictionless                                                                                                         |

|                  |                                                                   |
|------------------|-------------------------------------------------------------------|
| Scalar vs vector | Scalars have magnitude only; vectors have magnitude and direction |
| $W = mg$         | Converts mass (scalar, kg) to weight (vector force, N)            |

**Common exam mistakes**

- Giving a vague, generic criticism of a model ("it's not realistic") instead of naming the specific assumption and its effect in context.
- Using "mass" and "weight" interchangeably, or mixing up their units (kg for mass, N for weight).
- Treating distance/speed (scalars) and displacement/velocity (vectors) as the same thing.
- Forgetting to convert given quantities into SI units before calculating.
- Skipping working when converting units — write out each step so that method marks are not lost even if the final answer is wrong.

**You should now be able to**

- State and apply standard modelling assumptions (particle, rod, light, inextensible, smooth, uniform), and explain what each one ignores.
- Comment critically on the appropriateness of a model for a given real-world scenario.
- Distinguish scalar and vector quantities, including distance vs displacement and speed vs velocity.
- Convert between units and use  $W = mg$  to relate mass and weight.